HiLTL System Architecture

Main Component
Processing Module
Data/State
Output
Pathway

1. Environment and Automaton Interaction

LTL Formula φ
LTL→LDBA Compiler
State q_t
Environment
LDBA Aφ
Observation o_t
Automaton-Env Interaction

Automaton Graph

2. SE-GNN Encoding

Automaton Graph with Semantic Features
GATv2 Layer 1
(Edge-conditioned Attention)
GATv2 Layer 2
GATv2 Layer 3
Encoded Graph Representation

Encoded Features

3. Dual-Pathway Value Decomposition

Encoded Features
Short-term Pathway
Long-term Pathway
Multi-scale Embedding
Automaton Tracking
V_short
V_long

4. Dynamic Fusion and Policy

Dynamic Fusion: β(q)×V_short + (1-β(q))×V_long
Q-values
Policy π(a|s)
Action a_t